

Assignment 2

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Deadline Monday 15 May 12:30 pm

You MUST work in groups (max 3 people).

Send me by email (giuseppe.cinquemani@unicatt.it) your work including your writings and your REPLICA-BLE codes (you must test it and be sure it works properly) in a zip file named “ex1 and a reference name”. Use short names otherwise the compressed file will not work properly. If the system of the unovesity does not allow you to send a zip file containing an R file for security reasons, send me a link to some cloud or drive site from which I can download your assignment solution.

Your writing must contain name and surname of all the authors at the beginning of each file in the zip file. Remember all the members of the working group MUST be able to answer about the code and the writings you send me, even if you did not contribute all in the same way. Those who can not justify or understand the content of your solution will obtain mark zero for that exercise.

Extra bonus will be assigned to those asked to present the results during the dedicated class.

1. Forecasting the term structure of interest rates

We use two related econometric approaches:

1. a **two-step dynamic Nelson–Siegel model**, where the yield curve is first summarized by level, slope, and curvature factors and the factors are then forecast by VAR or AR models;
2. a **state-space dynamic Nelson–Siegel model**, where the same level, slope, and curvature factors are treated as latent states and estimated by the Kalman filter.

Question 1. Data preparation and descriptive analysis

Download daily Treasury yields from FRED and convert them to monthly averages. Keep the sample after removing missing observations.

Report the starting date, ending date, number of monthly observations, and maturities used.

Plot the yield curve at three dates of your choice: one early date, one middle date, and one recent date.

Compute and plot three empirical yield-curve summaries:

$$\text{Level}_t = y_t(120)$$

$$\text{Slope}_t = y_t(120) - y_t(3)$$

$$\text{Curvature}_t = 2y_t(24) - y_t(3) - y_t(120)$$

Briefly comment on the persistence and cyclical behavior of these three series.

Question 2. Nelson–Siegel cross-sectional factor extraction

For each month t , estimate the Nelson–Siegel curve

$$y_t(\tau) = \beta_{1t} + \beta_{2t} \left(\frac{1 - \exp(-\lambda\tau)}{\lambda\tau} \right) + \beta_{3t} \left[\frac{1 - \exp(-\lambda\tau)}{\lambda\tau} - \exp(-\lambda\tau) \right] + e_t(\tau)$$

where τ is maturity in months.

Use

$$\lambda = 0.0609$$

Tasks:

1. Construct the Nelson–Siegel loading matrix.
2. Estimate β_{1t} , β_{2t} , and β_{3t} month by month by OLS.
3. Plot the three estimated factors.
4. Compare them with empirical level, slope, and curvature.
5. Compute average fitting errors by maturity.

Question 3. VAR model for the Nelson–Siegel factors

Let

$$f_t = \begin{bmatrix} \beta_{1t} \\ \beta_{2t} \\ \beta_{3t} \end{bmatrix}$$

Estimate a VAR model for the Nelson–Siegel factors:

$$f_t = c + A_1 f_{t-1} + \cdots + A_p f_{t-p} + u_t$$

Produce recursive out-of-sample forecasts for

$$h = 1, \quad h = 6, \quad h = 12$$

Convert the factor forecasts into yield forecasts by

$$\hat{y}_{t+h|t}(\tau) = \hat{\beta}_{1,t+h|t} + \hat{\beta}_{2,t+h|t} \left(\frac{1 - \exp(-\lambda\tau)}{\lambda\tau} \right) + \hat{\beta}_{3,t+h|t} \left[\frac{1 - \exp(-\lambda\tau)}{\lambda\tau} - \exp(-\lambda\tau) \right]$$

Compare the VAR forecasts with the random-walk benchmark

$$\hat{y}_{t+h|t}(\tau) = y_t(\tau)$$

Report MAE and RMSE by maturity and forecast horizon.

Question 4. Kalman-filter state-space model

Estimate or implement a dynamic Nelson–Siegel state-space model.

The measurement equation is

$$y_t = \Lambda(\lambda)f_t + \varepsilon_t$$

where y_t is the vector of observed yields across maturities, $\Lambda(\lambda)$ is the Nelson–Siegel loading matrix, and

$$\varepsilon_t \sim N(0, H)$$

The transition equation is

$$f_t - \mu = A(f_{t-1} - \mu) + \eta_t$$

where

$$\eta_t \sim N(0, Q)$$

Tasks:

1. Explain the difference between the two-step OLS approach and the one-step Kalman-filter approach.
2. Estimate the model with diagonal Q and diagonal H .
3. Plot the filtered level, slope, and curvature factors.
4. Produce recursive forecasts for $h = 1, 6, 12$.
5. Compare the Kalman-filter forecasts with the VAR and random-walk forecasts.

Question 5. Interpretation and discussion:

Write a short discussion addressing the following questions:

1. Which model forecasts better at short horizons?
2. Which model forecasts better at long horizons?
3. Are the forecast gains concentrated at short, medium, or long maturities?
4. Does the Kalman-filter model produce smoother factors than the two-step OLS factors?
5. Why can a low-dimensional yield-curve factor structure improve forecasting performance even if the model is not arbitrage-free?